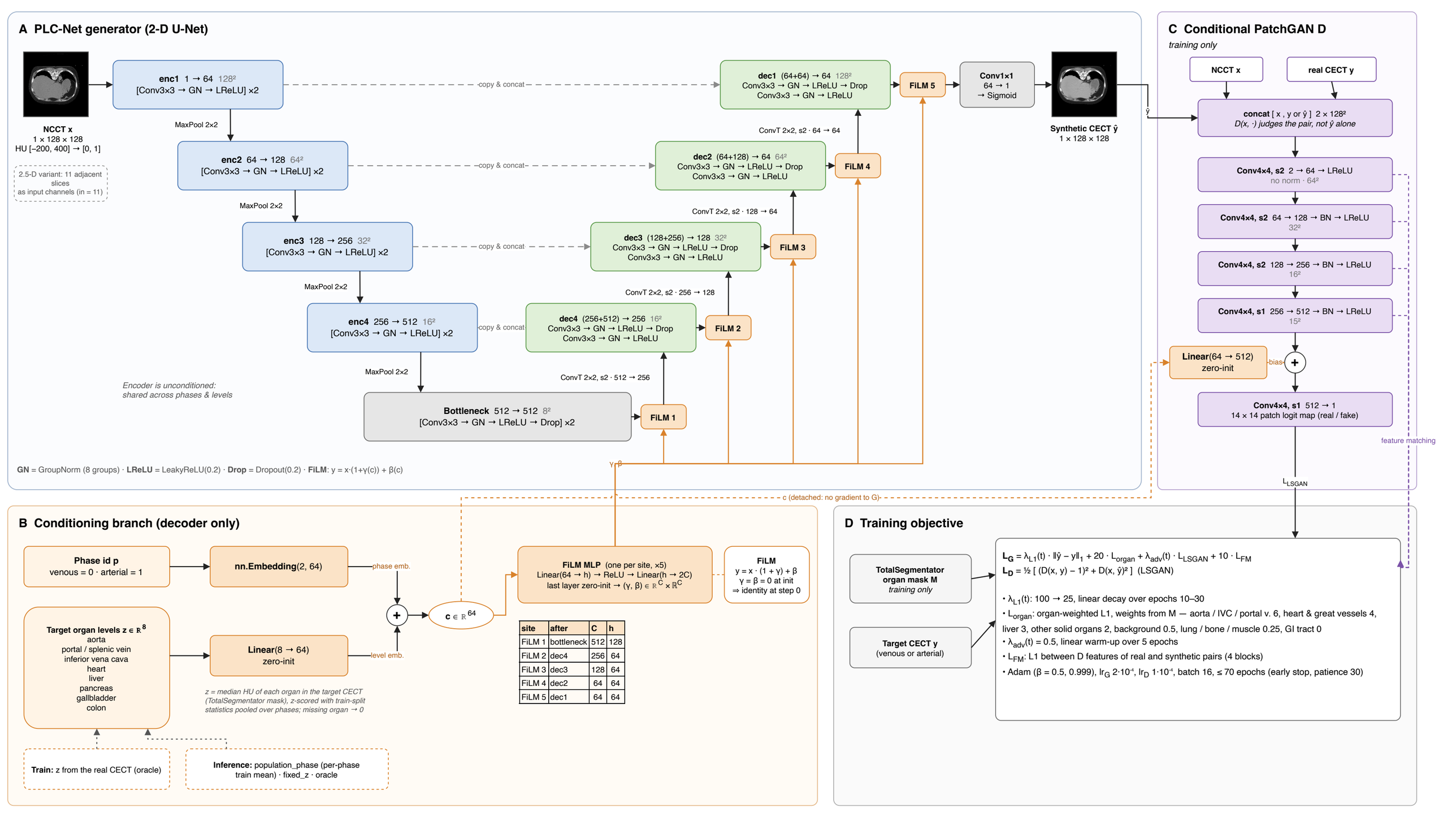

A PLC-Net generator (2-D U-Net)
NCCT x
1 × 128 × 128
HU [–200, 400] → [0, 1]
2.5-D variant: 11 adjacent slices as input channels (in = 11)
enc1 1 → 64 128²
[Conv3×3 → GN → LReLU] ×2
MaxPool 2×2
enc2 64 → 128 64²
[Conv3×3 → GN → LReLU] ×2
MaxPool 2×2
enc3 128 → 256 32²
[Conv3×3 → GN → LReLU] ×2
MaxPool 2×2
enc4 256 → 512 16²
[Conv3×3 → GN → LReLU] ×2
MaxPool 2×2
Bottleneck 512 → 512 8²
[Conv3×3 → GN → LReLU → Drop] ×2
Encoder is unconditioned: shared across phases & levels
copy & concat
dec1 (64+64) → 64 128²
Conv3×3 → GN → LReLU → Drop
Conv3×3 → GN → LReLU
ConvT 2×2, s2 · 64 → 64
dec2 (64+128) → 64 64²
Conv3×3 → GN → LReLU → Drop
Conv3×3 → GN → LReLU
ConvT 2×2, s2 · 128 → 64
dec3 (128+256) → 128 32²
Conv3×3 → GN → LReLU → Drop
Conv3×3 → GN → LReLU
ConvT 2×2, s2 · 256 → 128
dec4 (256+512) → 256 16²
Conv3×3 → GN → LReLU → Drop
Conv3×3 → GN → LReLU
ConvT 2×2, s2 · 512 → 256
FiLM 1
FiLM 2
FiLM 3
FiLM 4
FiLM 5
γ, β
Conv1×1
64 → 1
→ Sigmoid
Synthetic CECT ŷ
1 × 128 × 128
GN = GroupNorm (8 groups) · LReLU = LeakyReLU(0.2) · Drop = Dropout(0.2) · FiLM: y = x·(1+γ(c)) + β(c)
C Conditional PatchGAN D
training only
NCCT x
real CECT y
concat [x , y or ŷ] 2 × 128²
D(x, ·) judges the pair, not ŷ alone
Conv4×4, s2 2 → 64 → LReLU
no norm · 64²
Conv4×4, s2 64 → 128 → BN → LReLU
32²
Conv4×4, s2 128 → 256 → BN → LReLU
16²
Conv4×4, s1 256 → 512 → BN → LReLU
15²
Linear(64 → 512)
zero-init
bias
Conv4×4, s1 512 → 1
14 × 14 patch logit map (real / fake)
feature matching
L_LSGAN
c (detached: no gradient to G)
B Conditioning branch (decoder only)
Phase id p
venous = 0 · arterial = 1
nn.Embedding(2, 64)
phase emb.
Target organ levels z ∈ ℝ⁸
aorta
portal / splenic vein
inferior vena cava
heart
liver
pancreas
gallbladder
colon
Linear(8 → 64)
zero-init
level emb.
c ∈ ℝ⁶⁴
z = median HU of each organ in the target CECT (TotalSegmentator mask), z-scored with train-split statistics pooled over phases; missing organ → 0
FiLM MLP (one per site, ×5)
Linear(64 → h) → ReLU → Linear(h → 2C)
last layer zero-init → (γ, β) ∈ ℝ^C × ℝ^C
FiLM
y = x · (1 + γ) + β
γ = β = 0 at init
⇒ identity at step 0
site | after | C | h
FiLM 1 | bottleneck | 512 | 128
FiLM 2 | dec4 | 256 | 64
FiLM 3 | dec3 | 128 | 64
FiLM 4 | dec2 | 64 | 64
FiLM 5 | dec1 | 64 | 64
Train: z from the real CECT (oracle)
Inference: population_phase (per-phase train mean) · fixed_z · oracle
D Training objective
TotalSegmentator organ mask M
training only
Target CECT y
(venous or arterial)
L_G = λ_L1(t) · ‖ŷ − y‖_1 + 20 · L_organ + λ_adv(t) · L_LSGAN + 10 · L_FM
L_D = ½ [(D(x, y) − 1)² + D(x, ŷ)²] (LSGAN)
• λ_L1(t): 100 → 25, linear decay over epochs 10–30
• L_organ: organ-weighted L1, weights from M — aorta / IVC / portal v. 6, heart & great vessels 4, liver 3, other solid organs 2, background 0.5, lung / bone / muscle 0.25, GI tract 0
• λ_adv(t) = 0.5, linear warm-up over 5 epochs
• L_FM: L1 between D features of real and synthetic pairs (4 blocks)
• Adam (β = 0.5, 0.999), lr_G 2·10⁻⁴, lr_D 1·10⁻⁴, batch 16, ≤ 70 epochs (early stop, patience 30)